PROJECT AND TEAM INFORMATION

Project Title

STUDENT ACADEMIC RESULT & COURSE REGISTRATION SYSTEM

Team Name: *Team #*	*CODE RUSHERS*
Team member 1 (Team Lead)	*Gairola, Anirudh - 2510370098* *2510370098@geu.ac.in* 
Team member 2	*Sain,Anjali - 2510011010* *2510011010@geu.ac.in* 
Team member 3	*Mukherjee,Pratham - 2510012243* *2510012243@geu.ac.in* 
Team member 4	*Saha,Bishesh - 2510382260* *2510382260@geu.ac.in* 

PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract

This system helps in managing the academic results and registering for the courses efficiently, easily, and quickly. In many colleges, there are several problems faced by the students when checking the results, registering for the courses, verifying the prerequisite conditions, and so on.Our aim with the development of this project is to develop a system which manages the academic record of the student as well as provides an effective course registration process. This system will enable the storage of the information of the student, courses, and results. The DSA concepts will enable in improving the searching and sorting of the records. The OOP concepts will allow the division of the program into modules. The system will check whether the student meets all the requirements for registration in the course. When the course becomes unavailable, the system can manage the waiting list of the students who wish to register for the course. The project makes use of concepts from both C and C++ programs. We have made this project with the objective of reducing the work of the people manually and increasing the efficiency in the operations.

Updated Project Approach and Architecture

We have adopted a modular architecture for developing our academic management system, where each function is separately developed and integrated together to form the complete system. We will be utilizing the C language for fundamental data structures and C++ for object-oriented programming approach which would make our system more easy to organize, understand, and maintain.Our system consists of modules for managing student records, managing course information, processing academic results, and course registration. Student and course information will be suitably arranged using data structures while searching and sorting algorithms can be employed for retrieving information from data structures. Student, course, and academic record data can be represented using classes and objects. When a student registers for a course, the system verifies if he meets the necessary prerequisites. In case the number of seats in a particular course is limited, then a waiting-list mechanism can be provided which would cater to those students unable to register immediately. The academic results of a student can be suitably stored, accessed, and displayed.As far as storing records for the current prototype is concerned, we can use local file system for storing information in between execution of program.

Tasks Completed

Task Completed	Team Member
Finalized the project idea,objective,and system requirements.	ANIRUDH GAIROLA
Planned the system modules for student records and course management.	ANJALI SAIN
Studied and selected suitable DSA concepts for searching and sorting records.	PRATHAM MUKHERJEE
Planned the academic result processing and course registration features.	BISHESH SAHA

Challenges/Roadblocks

One of the biggest problems we have to face as our team is how to integrate C and C++ concepts with the proper modularization of our application. Choosing appropriate data structures to store student records and writing efficient search and sort functions would be an important task as well. The other problem we would be facing is related to the design of accurate prerequisite checking and waiting list for limited capacity courses. In order to overcome the above mentioned problems, we have planned to divide tasks among team members, brainstorm over the solutions and practice the programming concepts we have to use. We will try to develop and test our modules individually and then integrate them to make our whole application functional.

Tasks Pending

Task Pending	Team Member (to complete the task)
Implement student record management and data structures.	PRATHAM MUKHERJEE
Develop course management and registration modules.	ANIRUDH GAIROLA
Implement searching,sorting,and file-handling features.	ANJALI SAIN
Develop result processing,prerequisite checks,and waiting list functionality.	BISHESH SAHA
Integrate all modules,test the complete system,and prepare the final report and presentation.	ALL MEMBERS

Project Outcome/Deliverables

The anticipated result of our project will be a centralised and organised system that simplifies the process of managing academic record and enrolling into classes. It will facilitate the systematic organisation of student data, course details, computation and display of academic scores as well as processing of enrolment requests. Our system will be designed in such a way that it makes use of appropriate data structures to organise records.

Prerequisite checking and course availability management is one of the important results we would like to achieve. This will facilitate easy registration by students in eligible courses and handling cases of insufficient class sizes. File handling will be used for permanent storing and accessing of records. The project will further demonstrate the application of C language programming, C++ concepts of object orientation as well as Data Structures and Algorithms in solving practical problems. The deliverables at the end of the project will be the source code of the program, the modules and testing documentation and a report on the project.

Progress Overview

The project has moved through the phases of planning and system designing. In this project, the idea of the project has been decided, and the objective of the project has been defined. The requirements necessary for the development of Student Academic Result and Course Registration System have been discussed.

The team has defined the importance of using C language for developing the data structures and C++ for developing the concept of object-oriented programming in the system. Searching, sorting, file handling, pre-requisite check, and waiting list have been considered as an essential part of the system.

Currently, the main focus of the team is in arranging the implementation process, and the programming concepts that should be used in the process. The preparation of the modules for the development process is also an important task that needs to be done.

The next phase in the development process will include the linking of modules, error handling, and testing the system as a whole.

Codebase Information

The project codebase will contain the source code written in C and C++, organized into modules for student records, course management, academic result processing, and course registration. The code will be maintained in a GitHub repository to support version control, collaboration, and tracking of changes. Important commits will include the initial project structure, module development, feature integration, bug fixes, and final testing.

Repository Link: https://github.com/sen339853-rgb/STUDENT-ACADEMIC-RESULT-COURSE-REGISTRATION-SYSTEM.git

Branch: Main

Key Commits: Initial setup, module development, and feature integration.

Testing and Validation Status

Test Type	**Status (Pass/Fail)**	**Notes**
Student Record Management	Pass	Testing record entry & retrieval
Course Management	Pass	Verify course details & availability
Searching and Sorting	Pass	Check record organization & retrieval
Prerequisite verification	Pending	Validate course eligibility
File Handling	Pending	Test saving & loading records
System Integration	Pending	Test all modules together

Deliverables Progress

Project requirements and modular system architecture have been defined. Selection of C and C++ programming languages as well as appropriate DSA and OOP concepts have also been completed. The planned deliverables are student record management, course management, processing of academic results, and registration of courses. Their development and integration are work in progress and/or scheduled for the next stage. prerequisite verification, waiting list management and file handling will be included to extend the system functionalities. Errors will be located by testing and debugging. The deliverables will be complete source code of a tested academic management system, project report and a presentation explaining the design of the system, features and outcomes.